West Science Nature and Technology
Vol. 3, No. 01, March 2025, pp. 1~6

AI-Powered Service Quality Management for Next-Generation Industrial IoT in Indonesia

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Article Info

Article history:

Received March, 2025 Revised March, 2025
Accepted March, 2025

Keywords: AI-powered service quality, Industrial Internet of Things (IIoT), qualitative analysis, Indonesia, NVIVO

ABSTRACT

This study explores the integration of AI-powered service quality management in next-generation Industrial Internet of Things (IIoT) environments in Indonesia using a qualitative research approach. Drawing insights from five informants across diverse industries, the study investigates and the impact of AI technologies. Findings reveal that AI significantly enhances service quality through real-time monitoring, predictive

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analytics, and automation, leading to improved operational efficiency and customer satisfaction. However, challenges such as inadequate infrastructure, skill gaps, and data privacy concerns hinder widespread adoption, especially for small and medium enterprises (SMEs). The study emphasizes the importance of digital infrastructure development, workforce upskilling, and robust data security frameworks to fully realize the benefits of AI in IIoT environments. These findings contribute valuable insights for policymakers, industry practitioners, and researchers, providing a foundation for addressing contextual challenges and optimizing AI-driven service quality management in Indonesia.

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1.INTRODUCTION

The Industrial Internet of Things (IIoT) is revolutionizing industrial operations through interconnected systems, real-time data monitoring, and automation, enhancing efficiency and service quality. In Indonesia, IIoT adoption is accelerating to support industrial growth and digital transformation, though maintaining consistent service quality remains a challenge. IIoT integrates smart devices with traditional infrastructure for data-driven decision-making [1], enabling

remote management to improve efficiency and reduce costs [1]. AI and AIoT enhance automation, real-time decision-making, and predictive maintenance, optimizing resource utilization [2], [3]. However, IIoT implementation faces challenges such as data processing complexities [4], security risks, scalability issues [5], and high costs requiring workforce reskilling [3]. Despite these obstacles, IIoT improves product quality and reduces defects [4], with case studies showing significant efficiency gains and cost reductions [5].

Artificial Intelligence (AI) enhances service quality management in the Industrial Internet of Things (IIoT), particularly in Indonesia, where industrial sectors face challenges like infrastructure limitations, skill gaps, and data security concerns. AI-powered tools analyze large disruptions, optimize processes, and enable proactive decision-making, productivity and competitiveness. AI-driven technologies, including machine learning and predictive analytics, improve process control, reduce variability, and enhance quality [3]. Predictive maintenance, monitoring, and process optimization are crucial for maintaining service quality in IIoT environments [3]. AI-powered monitoring systems enhance data processing through automation and anomaly detection, ensuring effective governance [6]. Predictive analytics decision-making, reducing downtime and optimizing strategies [7]. AI-IoT integration in predictive maintenance minimizes unplanned downtime and optimizes schedules, lowering costs and improving reliability [14]. AI-driven quality control automates defect detection and monitoring, impr accuracy, cost efficiency, and consistency, essential for high service quality in IIoT environments [16].

Despite its implementation of AI-driven service quality management in Indonesia's IIoT landscape remains underexplored. Existing studies predominantly focus c advancements or sector-specific applications without addressing the broader implications for service quality. This gap highlights the need for a qualitative exploration of how AI can transform service quality management practices within the particularly in the Indonesian context. This study aims to fill this gap by investigating the role of AI-powered tools in managing and enhancing service quality in next-generation IIoT systems in Indonesia.

2.LITERATURE REVIEW

2.1 Service Quality Management in Industrial Environments

The integration of Artificial Intelligence (AI) in manufacturing enhances process control and efficiency by enabling dynamic adaptation, reducing variability, and improving quality. AI technologies such as machine learning, predictive analytics, and robotics drive productivity, cost-effectiveness, and sustainability in a competitive market. AI-driven solutions like predictive maintenance and real-time monitoring improve operational efficiency by detecting failures early and streamlining workflows [3], [8]. AI automation reduces human error and supports scalability, ensuring high-quality standards [9]. Additionally, AI optimizes resource utilization, minimizes waste, and improves material flows in closed-loop manufacturing [10], while AI-driven recycling strategies enhance material recovery and reduce raw material consumption [10].

2.2 The Role of Artificial Intelligence in Service Quality Management

AI has transformed service quality management by enhancing efficiency, predictive maintenance, and real-time data analysis, particularly in industrial and IIoT ecosystems. It optimizes processes, identifies **bottlenecks, and ensures BM quantum computers leverage furniture standards,** though adoption faces challenges like high costs, skill gaps, and data privacy concerns. AI-driven machine learning and predictive analytics enable real-time monitoring and process optimization [3]. In semiconductor manufacturing, predictive maintenance using IoT sensors reduces downtime and improves product quality [11]. However, implementation costs and workforce reskilling remain barriers [3], while data privacy requires strong governance [12]. AI also boosts business efficiency by automating data processing [13] and enhances defect detection in automotive manufacturing, lowering costs and improving customer satisfaction [14].

2.3 The Evolution of Industrial Internet of Things (IIoT) The Industrial Internet of Things (IIoT) has transformed industrial operations

IIoT, there is limited research focusing on their application in service quality

management, particularly in the Indonesian context. Most studies prioritize quantitative by integrating interconnected devices, evaluations of system performance without sensors, and systems to enhance efficiency considering the qualitative aspects of and productivity, driven by the convergence implementation, such as organizational of IoT with AI, machine learning, and cloud computing for real-time data analysis and

readiness, user perception, and contextual challenges. This study addresses this gap by
decision-making. However, maintaining employing a
service quality in IIoT environments presents
challenges, including data interoperability,

how AI-powered tools can be effectively utilized to enhance service quality in Indonesia's next-generation IIoT infrastructure.

integration of diverse devices often causes compatibility issues, hindering seamless data exchange [4], [15], while latency disrupts real-time processing in time-sensitive operations [2]. Additionally, the vast data generated by IIoT requires robust storage and processing infrastructure, posing a significant challenge [16]. AI plays a crucial role in addressing these issues by enabling real-time decision-making to reduce downtime and optimize resource utilization [2], predictive maintenance to prevent equipment failures [4], and process optimization to

latency, and environments.

efficiency [16]. Strategies like edge computing help reduce latency by processing data closer to the source [2], while standardization efforts improve data interoperability across systems [15].

2.4 Theoretical Framework

This study draws on the Technology-Organization-Environment (TOE) framework to examine the factors influencing the adoption of AI-powered service quality management in IIoT. The TOE framework considers technological

organizational readiness, and environmental factors as key determinants of technology adoption (Tornatzky & Fleischer, 1990). By applying this framework, the study identifies the interplay betw

advancements, organizational dynamics, and external pressures in shaping service quality management practices in Indonesia's IIoT sectors.

2.5 Research Gap

While existing literature extensively explores the technical capabilities of AI and

3.METHODS

3.1 Research Design

A qualitative approach was chosen for its ability to provide a nuanced understanding of the complex and dynamic interactions between AI technologies and service quality management in IIoT settings. This approach allows for the exploration of subjective perspectives, contextual factors, and emerging themes that may not be captured through quantitative methods. The study utilizes a case study design, focusing on selected informants from key industrial sectors in Indonesia, to investigate real-world applications and challenges.

3.2 Sampling Strategy

The study employs a purposive sampling technique to select participants with relevant expertise in AI, IIoT, and service quality management, ensuring informed insights into the research topic. The inclusion criteria require informants to have at least five years of professional experience in industrial operations, technology integration, or service management, direct involvement in AI or IIoT-related projects, and familiarity with the challenges and opportunities of adopting advanced technologies in Indonesia's industrial sectors. A total of five informants were selected from diverse industries, including manufacturing, logistics, and technology services, aligning with the study's qualitative approach to ensure depth and detail in the data collected.

3.3 Data Collection

Data were collected through semi-

shortens the lag time in problem identification and enables us to respond faster than before."

structured interviews, allowing flexibility in exploring topics while maintaining a focus on the study's objectives. The interview guide was developed based on key themes from the literature review, including perceptions of

However, concerns about the astronomical installation and maintenance costs were raised quite frequently, AI's role in service quality management, challenges in integrating AI with IIoT

particularly by SME informants.

4.2 Challenges in Implementing AI with IIoT

technologies, organizational

driven solutions, and the impact of AI on operational efficiency and service quality. Each interview lasted approximately 60

interviews:

There were some challenges faced during the

- The informants pointed out that

minutes and was conducted in

person or via

substandard digital infrastructure in the majority of Indonesian regions hinders the convergence of AI with IIoT without impediments.

- Lack of

video conferencing, depending on the informants' availability. With participants' consent, interviews were audio-recorded and

among employees was discovered to be one of the most significant impediments to adopting AI-powered solutions.

- The informants pointed out the need for robust data protection systems to protect sensitive industrial

transcribed verbatim for analysis.

3.4 Data Analysis

The data were analyze using NVIVO software, a qualitative analysis tool that facilitates systematic coding and exploration. The process began with data familiarization, where interview transcripts

information from infringements. One respondent said:

"Wh

were read multiple times to gain a comprehensive understanding. Key phrases and text segments were then coded based on recurring themes, patterns, and concepts. These codes were grouped into broader themes representing the study's core findings. Finally, the identified themes were analyzed in relation to the research objectives and

the unavailability of experts and a sound infrastructure makes it challenging to maximize such technologies."

4.3 Organizational Readiness

Organizational readiness to embrace AI-based management of service quality was also much dissimilar. Large organizations were better equipped environments. They emphasized the capability of AI to conduct real-time monitoring, predictive analysis, and automated decision-making as major drivers of improved opera

existing literature to draw meaningful conclusions.

4.RESULTS AND DISCUSSION

4.1 Perceptions of AI in Service Quality Management

All informants shared a common perception of the revolutionary nature of AI to improve service quality in IIoT

According to one informant:

"AI has revolutionized the manner in which we monitor and manage service quality. It

and superior infrastructure available. It was challenging to organize the right investment and brains for SMEs. Leadership commitment was also noted by the informants as being the key for successful implementation of AI.

4.4 Impact on Operational Efficiency and Service Quality

Despite the challenges, informants reported measurable improvements in operational efficiency and service quality where AI-powered tools had been implemented. Key benefits included reduced downtime, enhanced predictive maintenance, and improved customer satisfaction. For instance, one informant shared:

"By BM quantum computers leverage furniture AI into our IIoT systems, *we've seen a 30% reduction in service*

interruptions and a noticeable improvement in client feedback."

DISCUSSION

The findings affirm existing literature that states AI potential in enhancing service

context-specific AI and IIoT integration methods in new economies. Cross-regional comparative analysis can be a possible research direction in the future, or research can analyze sector-specific challenges and opportunities to create more tailored implementation plans.

5.CONCLUSION

quality management [18], particularly in real-time decision-making. Challenges such as skill shortages and infrastructure limitation, however, pose specific context-based challenges in Indonesia, according to [19], [20]. Challenges necessitate context-based strategies to ensure efficient AI implementation in industrial processes.

Solutions to these problems will require public-private Investment in digital infrastructure like the rollout of high-speed internet and investment in data centers will be required to provide enabling conditions for AI-driven solutions. Upgrading employee skills in AI and IIoT technologies through training programs will also be essential to bridge the skill gap and allow for easier integration. While the prohibitively high cost of adopting AI is a hurdle, long-term returns on operational productivity and service quality are worth the investment. Theoretical funding models, e.g., public-private partnership models, could guarantee access by SMEs to AI-based solutions and remain competitive.

Further, robust sound data security measures are a must as industries rely more on inter-connected systems. Protection from cyber threats to sensitive information has to be ensured, and that requires stringent regulatory measures in the form of data protection legislation. The research also focuses on undertaking further research into

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The study highlights the innovative potential of AI-based service quality management for next-generation IIoT platforms in Indonesia, where AI can propel service quality, operational effectiveness, and customer satisfaction. Adoption remains hindered by infrastructural limitations, skill gaps, and data privacy. Among some of the key findings are that while bigger firms are more likely to integrate AI, SMEs face challenges in availability of resources and technical skills. These challenges demand collaboration from institutions, the private sector, and government to invest in training, build robust data security, and improve digital infrastructure.

In order to ensure equitable benefits across industries, certain policies and finance models have to be adopted to enable the uptake of AI-based solutions for SMEs. Future research has to consider sector-specific strategies and comparative analysis to drive optimal embedding of AI in IIoT ecosystems. By overcoming these issues and seizing AI-based opportunities, Indonesia can unlock the full potential of AI-based service quality management and increase its industrial competitiveness in the global economy.

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